Manual Calculation Worksheets

Worked steps for the individual component. Copy your two files by hand, scan the sheets and attach them to Appendix A. Line numbers refer to Python-Markdown at commit `0bf535bc`.

Repository: https://github.com/Taha9876/SRE-A1

Member	File 1	File 2
Taha Yasir (i222429)	`markdown/extensions/def_list.py`	`markdown/extensions/meta.py`
Khizar (i222595)	`markdown/extensions/admonition.py`	`markdown/extensions/wikilinks.py`
Hadeed Abbas (i228780)	`markdown/postprocessors.py`	`markdown/extensions/legacy_attrs.py`

Rules to write at the top of your sheet

1. Comment density (SonarQube definition)

```
Comment density = comment lines / (comment lines + code lines) x 100

blank line     : contains only whitespace (not counted anywhere)
comment line   : a line that is only a # comment, or any non-blank line of a docstring
code line      : every other non-blank line (code + trailing # comment = code line)
```

2. Cyclomatic complexity (McCabe, per function)

```
CC = D + 1        D = number of decision points in the function

+1 for each:  if, elif, for, while, except, conditional expression (a if c else b),
              each `and` / `or`, each `for` and each `if` inside a comprehension
+0 for:       else, try, finally, return, break
A nested function is its own unit with its own +1.     File CC = sum over all functions.
```

3. Cognitive complexity (SonarSource, G. A. Campbell)

```
Start at 0 for each function.
Structural increment  +1 : if, elif, else, for, while, except, conditional expression,
                           each sequence of the same boolean operator (a and b and c = +1)
Nesting increment     +N : added to if, for, while, except, conditional expression.
                           N = number of enclosing if/elif/else, loop, except
                           or nested-function blocks.
                           elif, else and boolean sequences get NO nesting increment.
Increment written in the tables below = 1 + N.
```

Taha Yasir (i222429) - File 1: markdown/extensions/def_list.py

Step 1. Comment density

Category	Line numbers	Count
Blank	3, 5, 8, 10, 12, 14, 17, 21, 23, 28-29, 32, 35, 38, 40, 70, 88, 91-92, 95, 101, 104, 107-108, 111, 115, 120, 125-126	29
# comment	1-2, 4, 6-7, 9, 11, 13, 57-58, 62, 66, 72, 77, 79, 83, 96, 122	18
Docstring	15-16, 18-20, 31, 94, 98, 100, 103, 110, 113-114, 116-119, 121	18
Code	22, 24-27, 30, 33-34, 36-37, 39, 41-56, 59-61, 63-65, 67-69, 71, 73-76, 78, 80-82, 84-87, 89-90, 93, 97, 99, 102, 105-106, 109, 112, 123-124, 127-128	63
Total	**check: 29 + 18 + 18 + 63**	**128**

Line(s) 127: code followed by a trailing # comment, counted as code.

```
Comment lines   = 18 + 18 = 36
Code lines      = 63
Comment density = 36 / (36 + 63) x 100 = 36 / 99 x 100 = 36.4 %

Cross-check (radon raw): SLOC = 63 (matches the code count), blank = 29
```

Step 2. Cyclomatic complexity

DefListProcessor.run (lines 39-90)

Line	Decision point	Add	Running CC
-	function entry	1	1
43	comprehension 'for'	+1	2
44	comprehension 'if' x1	+1	3
47	if / elif	+1	4
51	if / elif	+1	5
56	if / elif	+1	6
56	boolean operator 'and' x1	+1	7
61	if / elif	+1	8
61	boolean operator 'and' x1	+1	9
71	if / elif	+1	10
71	boolean operator 'and' x1	+1	11
74	if / elif	+1	12
74	boolean operator 'and' x3	+3	15
80	for loop	+1	16
89	if / elif	+1	17
	CC = 16 decision points + 1		**17**

Tool cross-check: radon reports 17. Match.

Functions with no decision point (CC = 1 each): `test`, `create_item`, `extendMarkdown`, `makeExtension`. That is 4 x 1 = 4.

```
File cyclomatic complexity = 17 + 4 = 21
Highest function = 17      Average = 21 / 5 functions = 4.20
```

Step 3. Cognitive complexity (stepwise)

DefListProcessor.run (lines 39-90)

Line	Construct	Nesting level	Increment	Add	Running total
47	if	0	+1	+1	1
49	else	-	+1	+1	2
51	if	0	+1	+1	3
53	else	-	+1	+1	4
56	if	0	+1	+1	5
56	sequence of 'and'	-	+1	+1	6
61	if	0	+1	+1	7
61	sequence of 'and'	-	+1	+1	8
68	else	-	+1	+1	9
71	if	0	+1	+1	10
71	sequence of 'and'	-	+1	+1	11
74	if	1	+1 +1 nesting	+2	13
74	sequence of 'and'	-	+1	+1	14
76	else	-	+1	+1	15
80	for loop	0	+1	+1	16
89	if	0	+1	+1	17
	Cognitive complexity				**17**

Tool cross-check: complexipy reports 19. See the note below.

Functions with a straight-line body (cognitive complexity 0): `test`, `create_item`, `extendMarkdown`, `makeExtension`.

```
File cognitive complexity = 17        Highest function = 17
```

Note. complexipy reports 19 for `DefListProcessor.run` because it also counts the list comprehension on lines 43-44 (+2). The SonarSource rules do not, so the hand result is 17. Write both numbers on your sheet.

Result to carry into Appendix A

Comment density	Cyclomatic (file / max)	Cognitive (file / max)
36.4 %	21 / 17	17 / 17

Taha Yasir (i222429) - File 2: markdown/extensions/meta.py

Step 1. Comment density

Category	Line numbers	Count
Blank	3, 5, 8, 10, 12, 14, 17, 21, 23, 29, 31, 37-38, 41, 44, 48, 54, 57-58, 61, 90-91	22
# comment	1-2, 4, 6-7, 9, 11, 13, 32, 50, 83	11
Docstring	15-16, 18-20, 40, 43, 45-47, 49, 60, 63	13
Code	22, 24-28, 30, 33-36, 39, 42, 51-53, 55-56, 59, 62, 64-82, 84-89, 92-93	47
Total	**check: 22 + 11 + 13 + 47**	**93**

Line(s) 72, 87, 92: code followed by a trailing # comment, counted as code.

```
Comment lines   = 11 + 13 = 24
Code lines      = 47
Comment density = 24 / (24 + 47) x 100 = 24 / 71 x 100 = 33.8 %

Cross-check (radon raw): SLOC = 47 (matches the code count), blank = 22
```

Step 2. Cyclomatic complexity

MetaPreprocessor.run (lines 62-89)

Line	Decision point	Add	Running CC
-	function entry	1	1
66	if / elif	+1	2
66	boolean operator 'and' x1	+1	3
68	while loop	+1	4
71	if / elif	+1	5
71	boolean operator 'or' x1	+1	6
73	if / elif	+1	7
78	except clause	+1	8
82	if / elif	+1	9
82	boolean operator 'and' x1	+1	10
	CC = 9 decision points + 1		**10**

Tool cross-check: radon reports 10. Match.

Functions with no decision point (CC = 1 each): `extendMarkdown`, `reset`, `makeExtension`. That is 3 x 1 = 3.

```
File cyclomatic complexity = 10 + 3 = 13
Highest function = 10      Average = 13 / 4 functions = 3.25
```

Step 3. Cognitive complexity (stepwise)

MetaPreprocessor.run (lines 62-89)

Line	Construct	Nesting level	Increment	Add	Running total
66	if	0	+1	+1	1
66	sequence of 'and'	-	+1	+1	2
68	while loop	0	+1	+1	3
71	if	1	+1 +1 nesting	+2	5
71	sequence of 'or'	-	+1	+1	6

Line	Construct	Nesting level	Increment	Add	Running total
73	if	1	+1 +1 nesting	+2	8
78	except	2	+1 +2 nesting	+3	11
80	else	-	+1	+1	12
82	if	2	+1 +2 nesting	+3	15
82	sequence of 'and'	-	+1	+1	16
85	else	-	+1	+1	17
	Cognitive complexity				**17**

Tool cross-check: complexipy reports 17. Match.

Functions with a straight-line body (cognitive complexity 0): `extendMarkdown`, `reset`, `makeExtension`.

```
File cognitive complexity = 17        Highest function = 17
```

Result to carry into Appendix A

Comment density	Cyclomatic (file / max)	Cognitive (file / max)
33.8 %	13 / 10	17 / 17

Khizar (i222595) - File 1: markdown/extensions/admonition.py

Step 1. Comment density

Category	Line numbers	Count
Blank	3, 5, 7, 10, 12, 14, 16-17, 21, 23, 27, 29, 35, 38-39, 42, 46, 50, 54, 56-57, 59, 64, 67, 69, 72, 75, 78, 80, 83, 91, 93, 106, 111, 119, 122, 128, 130, 132, 137, 141, 149, 165, 167, 169, 175, 189-190	48
# comment	1-2, 4, 6, 8-9, 11, 13, 15, 52, 84, 97-98, 107-108, 112-114, 159, 171-173, 180-182, 185-186	27
Docstring	18-20, 22, 24-26, 41, 44-45, 47-49, 51, 66, 74, 76-77, 79	19
Code	28, 30-34, 36-37, 40, 43, 53, 55, 58, 60-63, 65, 68, 70-71, 73, 81-82, 85-90, 92, 94-96, 99-105, 109-110, 115-118, 120-121, 123-127, 129, 131, 133-136, 138-140, 142-148, 150-158, 160-164, 166, 168, 170, 174, 176-179, 183-184, 187-188, 191-192	98
Total	**check: 48 + 27 + 19 + 98**	**192**

Line(s) 36, 145, 191: code followed by a trailing # comment, counted as code.

```
Comment lines   = 27 + 19 = 46
Code lines      = 98
Comment density = 46 / (46 + 98) x 100 = 46 / 144 x 100 = 31.9 %

Cross-check (radon raw): SLOC = 98 (matches the code count), blank = 48
```

Step 2. Cyclomatic complexity

AdmonitionProcessor.parse_content (lines 73-129)

Line	Decision point	Add	Running CC
-	function entry	1	1
85	if / elif	+1	2
94	if / elif	+1	3
94	boolean operator 'or' x2	+2	5
101	while loop	+1	6
102	if / elif	+1	7
103	boolean operator 'and' x3	+3	10
110	conditional expression (x if c else y)	+1	11
120	if / elif	+1	12
123	if / elif	+1	13
	CC = 12 decision points + 1		**13**

Tool cross-check: radon reports 13. Match.

AdmonitionProcessor.test (lines 131-136)

Line	Decision point	Add	Running CC
-	function entry	1	1
133	if / elif	+1	2
	CC = 1 decision points + 1		**2**

Tool cross-check: radon reports 2. Match.

AdmonitionProcessor.run (lines 138-174)

Line	Decision point	Add	Running CC
-	function entry	1	1
142	if / elif	+1	2

Line	Decision point	Add	Running CC
143	if / elif	+1	3
150	if / elif	+1	4
154	if / elif	+1	5
160	if / elif	+1	6
160	boolean operator 'and' x1	+1	7
170	if / elif	+1	8
	CC = 7 decision points + 1		**8**

Tool cross-check: radon reports 8. Match.

AdmonitionProcessor.get_class_and_title (lines 176-188)

Line	Decision point	Add	Running CC
-	function entry	1	1
179	if / elif	+1	2
184	if / elif	+1	3
	CC = 2 decision points + 1		**3**

Tool cross-check: radon reports 3. Match.

Functions with no decision point (CC = 1 each): `extendMarkdown`, `__init__`, `makeExtension`. That is 3 x 1 = 3.

```
File cyclomatic complexity = 13 + 2 + 8 + 3 + 3 = 29
Highest function = 13      Average = 29 / 7 functions = 4.14
```

Step 3. Cognitive complexity (stepwise)

AdmonitionProcessor.parse_content (lines 73-129)

Line	Construct	Nesting level	Increment	Add	Running total
85	if	0	+1	+1	1
94	if	0	+1	+1	2
94	sequence of 'or'	-	+1	+1	3
96	else	-	+1	+1	4
101	while loop	1	+1 +1 nesting	+2	6
102	if	2	+1 +2 nesting	+3	9
103	sequence of 'and'	-	+1	+1	10
110	conditional expression	3	+1 +3 nesting	+4	14
117	else	-	+1	+1	15
120	if	1	+1 +1 nesting	+2	17
123	if	1	+1 +1 nesting	+2	19
	Cognitive complexity				**19**

Tool cross-check: complexipy reports 19. Match.

AdmonitionProcessor.test (lines 131-136)

Line	Construct	Nesting level	Increment	Add	Running total
133	if	0	+1	+1	1
135	else	-	+1	+1	2
	Cognitive complexity				**2**

Tool cross-check: complexipy reports 2. Match.

AdmonitionProcessor.run (lines 138-174)

Line	Construct	Nesting level	Increment	Add	Running total
142	if	0	+1	+1	1
143	if	1	+1 +1 nesting	+2	3
147	else	-	+1	+1	4
150	if	0	+1	+1	5
154	if	1	+1 +1 nesting	+2	7
158	else	-	+1	+1	8
160	if	1	+1 +1 nesting	+2	10
160	sequence of 'and'	-	+1	+1	11
170	if	0	+1	+1	12
	Cognitive complexity				**12**

Tool cross-check: complexipy reports 12. Match.

AdmonitionProcessor.get_class_and_title (lines 176-188)

Line	Construct	Nesting level	Increment	Add	Running total
179	if	0	+1	+1	1
184	elif	-	+1	+1	2
	Cognitive complexity				**2**

Tool cross-check: complexipy reports 2. Match.

Functions with a straight-line body (cognitive complexity 0): `extendMarkdown`, `__init__`, `makeExtension`.

```
File cognitive complexity = 19 + 2 + 12 + 2 = 35        Highest function = 19
```

Result to carry into Appendix A

Comment density	Cyclomatic (file / max)	Cognitive (file / max)
31.9 %	29 / 13	35 / 19

Khizar (i222595) - File 2: markdown/extensions/wikilinks.py

Step 1. Comment density

Category	Line numbers	Count
Blank	3, 5, 8, 10, 12, 14, 17, 21, 23, 29-30, 35-36, 39, 49, 52, 56, 60, 66-67, 70, 74, 88, 102-103	25
# comment	1-2, 4, 6-7, 9, 11, 13, 58, 61	10
Docstring	15-16, 18-20, 32, 38, 47, 51, 53-55, 57, 69, 90	15
Code	22, 24-28, 31, 33-34, 37, 40-46, 48, 50, 59, 62-65, 68, 71-73, 75-87, 89, 91-101, 104-105	55
Total	**check: 25 + 10 + 15 + 55**	**105**

Line(s) 104: code followed by a trailing # comment, counted as code.

```
Comment lines   = 10 + 15 = 25
Code lines      = 55
Comment density = 25 / (25 + 55) x 100 = 25 / 80 x 100 = 31.2 %

Cross-check (radon raw): SLOC = 55 (matches the code count), blank = 25
```

Step 2. Cyclomatic complexity

WikiLinksInlineProcessor.handleMatch (lines 75-87)

Line	Decision point	Add	Running CC
-	function entry	1	1
76	if / elif	+1	2
83	if / elif	+1	3
	CC = 2 decision points + 1		**3**

Tool cross-check: radon reports 3. Match.

WikiLinksInlineProcessor._getMeta (lines 89-101)

Line	Decision point	Add	Running CC
-	function entry	1	1
94	if / elif	+1	2
95	if / elif	+1	3
97	if / elif	+1	4
99	if / elif	+1	5
	CC = 4 decision points + 1		**5**

Tool cross-check: radon reports 5. Match.

Functions with no decision point (CC = 1 each): `build_url`, `__init__`, `extendMarkdown`, `__init__`, `makeExtension`. That is 5 x 1 = 5.

```
File cyclomatic complexity = 3 + 5 + 5 = 13
Highest function = 5      Average = 13 / 7 functions = 1.86
```

Step 3. Cognitive complexity (stepwise)

WikiLinksInlineProcessor.handleMatch (lines 75-87)

Line	Construct	Nesting level	Increment	Add	Running total
76	if	0	+1	+1	1

Line	Construct	Nesting level	Increment	Add	Running total
83	if	1	+1 +1 nesting	+2	3
85	else	-	+1	+1	4
	Cognitive complexity				**4**

Tool cross-check: complexipy reports 4. Match.

WikiLinksInlineProcessor._getMeta (lines 89-101)

Line	Construct	Nesting level	Increment	Add	Running total
94	if	0	+1	+1	1
95	if	1	+1 +1 nesting	+2	3
97	if	1	+1 +1 nesting	+2	5
99	if	1	+1 +1 nesting	+2	7
	Cognitive complexity				**7**

Tool cross-check: complexipy reports 7. Match.

Functions with a straight-line body (cognitive complexity 0): `build_url`, `__init__`, `extendMarkdown`, `__init__`, `makeExtension`.

```
File cognitive complexity = 4 + 7 = 11        Highest function = 7
```

Result to carry into Appendix A

Comment density	Cyclomatic (file / max)	Cognitive (file / max)
31.2 %	13 / 5	11 / 7

Hadeed Abbas (i228780) - File 1: markdown/postprocessors.py

Step 1. Comment density

Category	Line numbers	Count
Blank	2, 4, 8, 13, 17, 19, 21, 26, 28, 30, 34, 37-38, 45-46, 50, 53, 55, 57, 63, 66-67, 70, 72, 87, 94, 104, 108-109, 112, 116-117, 124, 126, 129	35
# comment	1, 3, 5-7, 9-12, 14-16, 18, 100	14
Docstring	20, 22-25, 27, 40, 48-49, 51-52, 54, 56, 59-62, 64, 69, 74, 96, 106, 111, 123	24
Code	29, 31-33, 35-36, 39, 41-44, 47, 58, 65, 68, 71, 73, 75-86, 88-93, 95, 97-99, 101-103, 105, 107, 110, 113-115, 118-122, 125, 127-128, 130-131	58
Total	**check: 35 + 14 + 24 + 58**	**131**

Line(s) 35, 65: code followed by a trailing # comment, counted as code.

```
Comment lines   = 14 + 24 = 38
Code lines      = 58
Comment density = 38 / (38 + 58) x 100 = 38 / 96 x 100 = 39.6 %

Cross-check (radon raw): SLOC = 58 (matches the code count), blank = 35
```

Step 2. Cyclomatic complexity

RawHtmlPostprocessor.run (lines 73-93)

Line	Decision point	Add	Running CC
-	function entry	1	1
88	if / elif	+1	2
	CC = 1 decision points + 1		**2**

Tool cross-check: radon reports 2. Match.

RawHtmlPostprocessor.run > nested function substitute_match (lines 75-86)

Line	Decision point	Add	Running CC
-	function entry	1	1
76	if / elif	+1	2
81	if / elif	+1	3
84	if / elif	+1	4
84	boolean operator 'or' x1	+1	5
	CC = 4 decision points + 1		**5**

Tool cross-check: radon reports 5. Match.

RawHtmlPostprocessor.isblocklevel (lines 95-103)

Line	Decision point	Add	Running CC
-	function entry	1	1
98	if / elif	+1	2
99	if / elif	+1	3
	CC = 2 decision points + 1		**3**

Tool cross-check: radon reports 3. Match.

Functions with no decision point (CC = 1 each): `build_postprocessors`, `run`, `stash_to_string`, `run`, `unescape`, `run`. That is 6 x 1 = 6.

```
File cyclomatic complexity = 2 + 5 + 3 + 6 = 16
Highest function = 5      Average = 16 / 9 functions = 1.78
```

Step 3. Cognitive complexity (stepwise)

RawHtmlPostprocessor.run (lines 73-93)

Line	Construct	Nesting level	Increment	Add	Running total
76	if	1	+1 +1 nesting	+2	2
78	else	-	+1	+1	3
81	if	1	+1 +1 nesting	+2	5
84	if	1	+1 +1 nesting	+2	7
84	sequence of 'or'	-	+1	+1	8
88	if	0	+1	+1	9
92	else	-	+1	+1	10
	Cognitive complexity				**10**

Tool cross-check: complexipy reports 10. Match.

RawHtmlPostprocessor.isblocklevel (lines 95-103)

Line	Construct	Nesting level	Increment	Add	Running total
98	if	0	+1	+1	1
99	if	1	+1 +1 nesting	+2	3
	Cognitive complexity				**3**

Tool cross-check: complexipy reports 3. Match.

Functions with a straight-line body (cognitive complexity 0): `build_postprocessors`, `run`, `stash_to_string`, `run`, `unescape`, `run`.

```
File cognitive complexity = 10 + 3 = 13        Highest function = 10
```

Note. The nested function `substitute_match` (lines 75-86) has its own +1 for cyclomatic complexity, but its cognitive complexity is charged to `run` one nesting level deeper, so line 76 scores 2.

Result to carry into Appendix A

Comment density	Cyclomatic (file / max)	Cognitive (file / max)
39.6 %	16 / 5	13 / 10

Hadeed Abbas (i228780) - File 2: markdown/extensions/legacy_attrs.py

Step 1. Comment density

Category	Line numbers	Count
Blank	2, 4, 8, 13, 17, 19, 22, 30, 32, 37, 40-41, 43-44, 56, 62-63, 67, 71, 75-76	21
# comment	1, 3, 5-7, 9-12, 14-16, 18, 73	14
Docstring	20-21, 23-29, 47, 58, 66, 68-70, 72	16
Code	31, 33-36, 38-39, 42, 45-46, 48-55, 57, 59-61, 64-65, 74, 77-78	27
Total	**check: 21 + 14 + 16 + 27**	**78**

Line(s) 38, 42, 77: code followed by a trailing # comment, counted as code.

```
Comment lines   = 14 + 16 = 30
Code lines      = 27
Comment density = 30 / (30 + 27) x 100 = 30 / 57 x 100 = 52.6 %

Cross-check (radon raw): SLOC = 27 (matches the code count), blank = 21
```

Step 2. Cyclomatic complexity

LegacyAttrs.run (lines 46-55)

Line	Decision point	Add	Running CC
-	function entry	1	1
48	for loop	+1	2
50	if / elif	+1	3
52	if / elif	+1	4
52	boolean operator 'and' x1	+1	5
54	if / elif	+1	6
54	boolean operator 'and' x1	+1	7
	CC = 6 decision points + 1		**7**

Tool cross-check: radon reports 7. Match.

Functions with no decision point (CC = 1 each): `handleAttributes`, `attributeCallback`, `extendMarkdown`, `makeExtension`. That is 4 x 1 = 4.

```
File cyclomatic complexity = 7 + 4 = 11
Highest function = 7      Average = 11 / 5 functions = 2.20
```

Step 3. Cognitive complexity (stepwise)

LegacyAttrs.run (lines 46-55)

Line	Construct	Nesting level	Increment	Add	Running total
48	for loop	0	+1	+1	1
50	if	1	+1 +1 nesting	+2	3
52	if	1	+1 +1 nesting	+2	5
52	sequence of 'and'	-	+1	+1	6
54	if	1	+1 +1 nesting	+2	8
54	sequence of 'and'	-	+1	+1	9
	Cognitive complexity				**9**

Tool cross-check: complexipy reports 9. Match.

Functions with a straight-line body (cognitive complexity 0): `handleAttributes`, `extendMarkdown`, `makeExtension`.

```
File cognitive complexity = 9        Highest function = 9
```

Note. The nested function `attributeCallback` (line 59) has no decision point: cyclomatic 1, cognitive 0.

Result to carry into Appendix A

Comment density	Cyclomatic (file / max)	Cognitive (file / max)
52.6 %	11 / 7	9 / 9